

Instructor Copy Lab 1

Name

2026-08-30

Details to walk through in class for students to do together

System Stuff and R Studio

- Create STA 209 directory
 - Hwk and labs
- What is R Studio and what is R
 - Remove save global data
 - Remove .Rdata
 - Different panes, assignments, viewer, console
 - * Create `x <- 4` and `y <- 6`
 - Open R Markdown, delete header shit, save to labs

R Markdown and friends

1. Create a new markdown document - Add your name and title - Delete the other shit
2. What are things we can do in markdown - Headers (correct and incorrect) - R Code chunks (see how we create things in environ) - Italics, bold, creating lists
3. How are we going to create homework solutions?

HAVE STUDENTS DO QUESTION 1 TOGETHER

R and friends

- Let's create a vector and play with functions

```
x <- c(1, 3, 5, 7, 9)
x^2
```

```
## [1] 1 9 25 49 81
```

```
sqrt(x)
```

```
## [1] 1.0000 1.7321 2.2361 2.6458 3.0000
```

```
mean(x)
```

```
## [1] 5
```

```
# ?sqrt
```

DO QUESTION 2

- We can use logic on stuff, creating categories

```
x < 4
```

```
## [1] TRUE TRUE FALSE FALSE FALSE
```

```
table(x < 4)
```

```
##
```

```
## FALSE TRUE
```

```
##      3      2
```

- R can be used to do all kinds of fun statistics things
 - What happens if I run it multiple times?
 - What happens if I change `prob` argument?
 - I can put this table in a barplot

```
## Collin stuff here
```

```
pets <- sample(c("Cat", "Dog", "Other"),  
              size = 30, replace = TRUE)
```

```
table(pets)
```

```
## pets
```

```
##   Cat   Dog Other
```

```
##    9    7    14
```

```
# barplot(table(pets))
```

```
pets <- sample(c("Cat", "Dog", "Other"),  
              size = 30, replace = TRUE, prob = c(0.1, 0.2, 0.7))
```

```
table(pets)
```

```
## pets
```

```
##   Cat   Dog Other
```

```
##    4    7    19
```

OKAY NOW HAVE THEM DO QUESTION 3

Data.frames and friends

First, we need to read in data

```
crawdad <- read.csv("https://collinn.github.io/data/crawdad.csv")
```

Then what can we learn:

- Env has a way to view
 - How many obs and vars, what are their types
 - Can also use `str()`
- I can use `$` to extract a variable. This is just a vector from before
 - Use `$` to get mean of a variable
 - Create a histogram with `hist()`
 - Use `$` to create table of categorical

NOW THEY CAN DO QUESTIONS 4-6